

Online Conformal Prediction with Efficiency Guarantees

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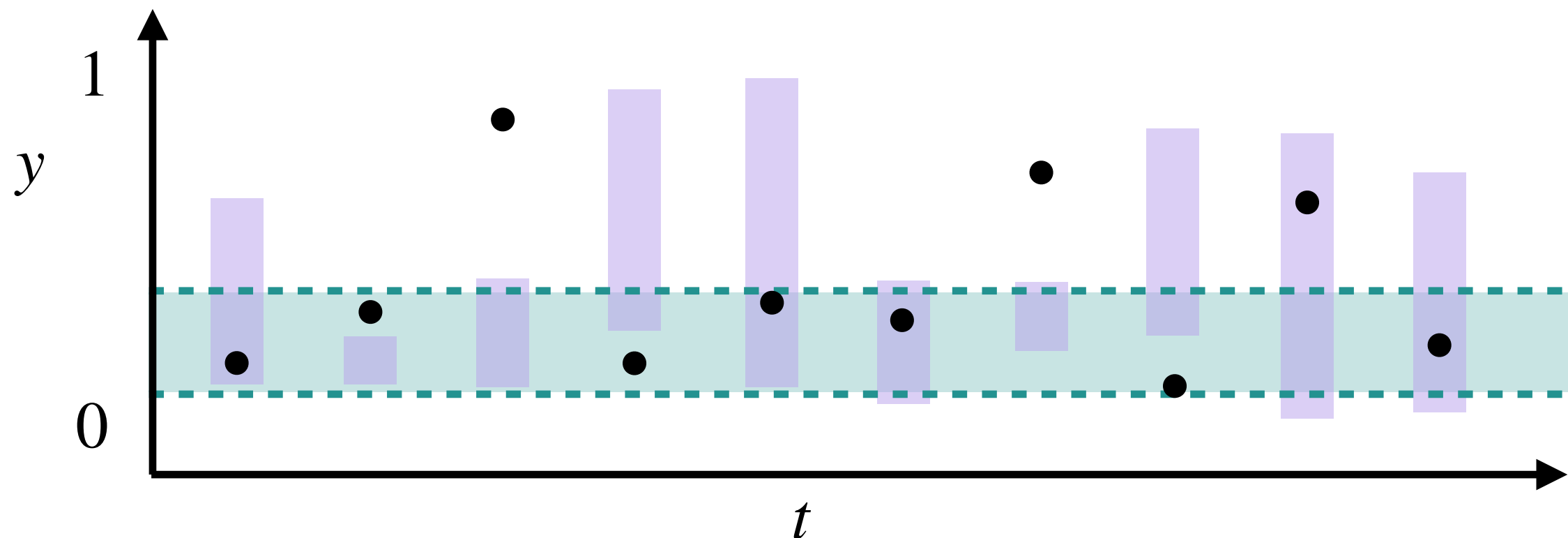
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Generating Prediction Sets Online

Predict a sequence of $y_t \in [0,1]$

For each day t , generate **prediction set** $C_t \subseteq [0,1]$,
to achieve **coverage** $1 - \alpha$ over y_t

Illustration: $1 - \alpha = 0.7$



Goals:

- (1) **Coverage:** capture (close to) $1 - \alpha$ fraction of the points
- (2) **Efficiency:** play average volume close to the **best fixed interval in hindsight** that achieves coverage $1 - \alpha$

Motivation: Conformal Prediction

Strategy to ensure **reliability** of machine learning in practice

Standard prediction problem:

See $(x_1, y_1), (x_2, y_2), \dots, (x_T, y_T)$, and x_{T+1} .

What is y_{T+1} ?

Learning theory answer:

Assume $(x_i, y_i) \sim \mathcal{D}$ i.i.d.. Then, for a reasonable hypothesis class $\mathcal{F}$, can learn function $f^* \in \mathcal{F}$, $\hat{y} = f^*(x)$, that has the lowest error on $\mathcal{D}$ (**regression**)

Issues in practice:

(1) (x_i, y_i) are not i.i.d.

→ consider **online model**

(2) If best function $f^* \in \mathcal{F}$ is bad (high error), want to know now!

Evaluating error at the end of the game is too late to do anything about it

Need **uncertainty quantification** on the fly → consider **prediction sets**

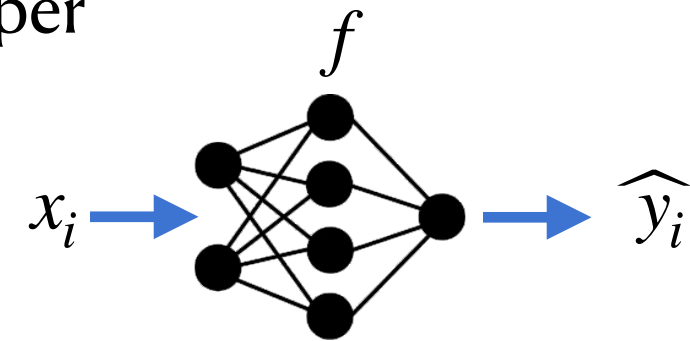
Conformal Prediction: Usual Recipe

“Wrap” **regression model** with conformal wrapper

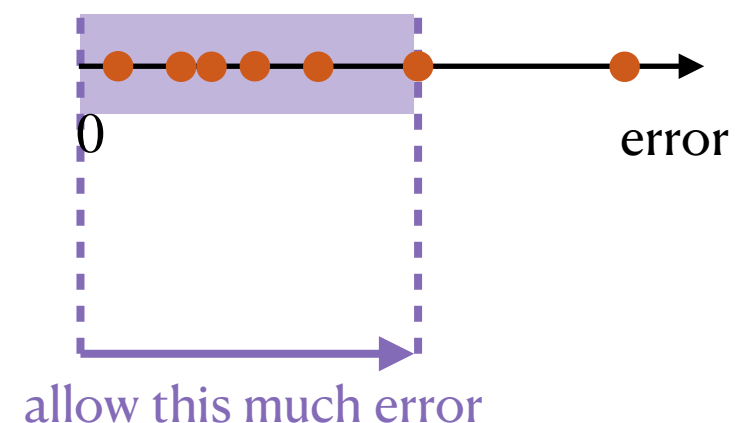
Strategy:

- (1) Train a regression model $f : X \rightarrow Y$ to predict $\hat{y}_i = f(x_i)$
- (2) Measure the error of the prediction according to a hand-chosen **non-conformity score** $s(y, \hat{y}) \in \mathbb{R}$
- (3) Estimate τ , the $(1 - \alpha)$ th quantile of the non-conformity score online
- (4) For new x_i , predict set of y that “conform” to error of f

$$C_i = \{y : s(\hat{y}_i, y) \leq \tau\}$$



$$s(y, \hat{y}) = \|\hat{y} - y\|$$



Pros:

- Ensure coverage with no guarantees required of f (could be a neural network)
- Simple recipe
- Wrapper **ensures reliability** of f

Cons:

- Need to design non-conformity score by hand for every new setting
- Efficiency (set size) depends a lot on non-conformity score!
→ no efficiency guarantees

Bottleneck: Set Size

Requires lots of work to design the non-conformity scores for new settings



The screenshot shows the arXiv search interface. At the top, there is a red header with the arXiv logo on the left. On the right side of the header, there is a search bar with the placeholder text 'Search...', a dropdown menu currently set to 'All fields', and a 'Search' button. Below the header, there is a navigation bar with links for 'Help' and 'Advanced Search', and a 'Logi' link on the far right. The main content area displays the text 'Showing 1-50 of 611 results for title: conformal prediction' in a large, bold font. To the right of this text, there is a smaller link that says 'Search v0.5.6 released 2020-'. Below this, there is a search bar with the text 'conformal prediction' entered, a dropdown menu set to 'Title', and a blue 'Search' button.

Large set sizes bottleneck conformal strategies in practice

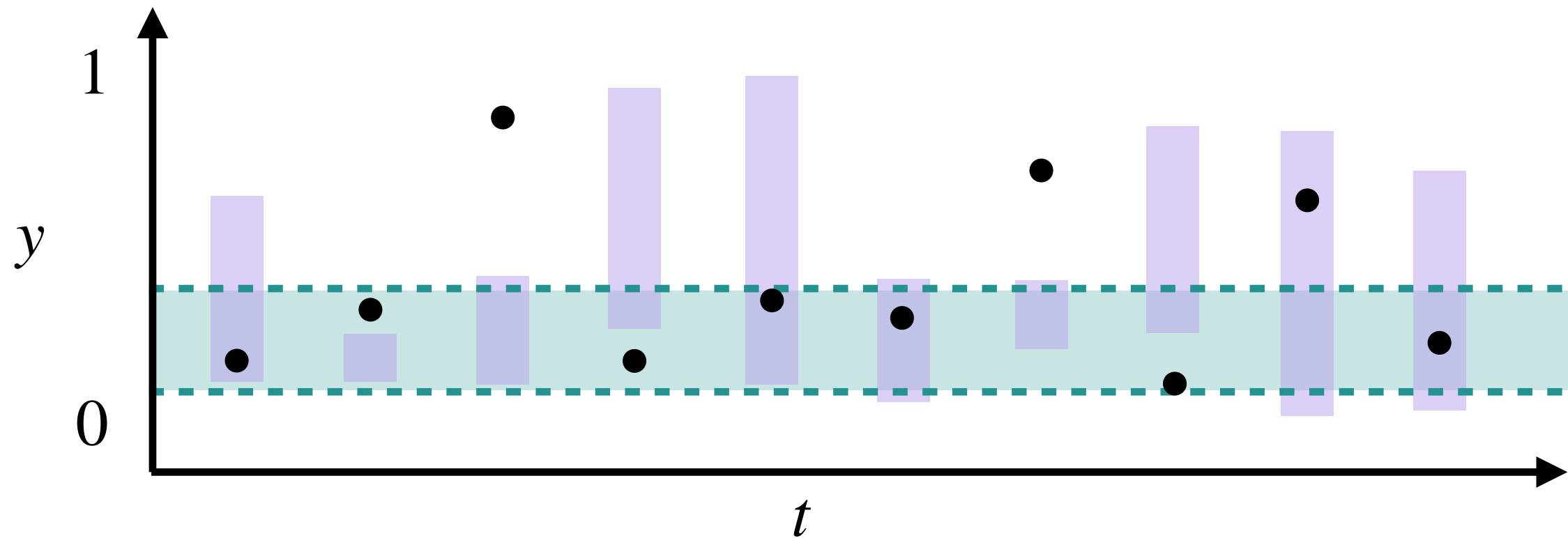
Questions:

- Can we automatically learn the smallest possible prediction sets?
- Is **regression** the right way to approach this problem?

This Work: Theoretical Study

Simplest setting:

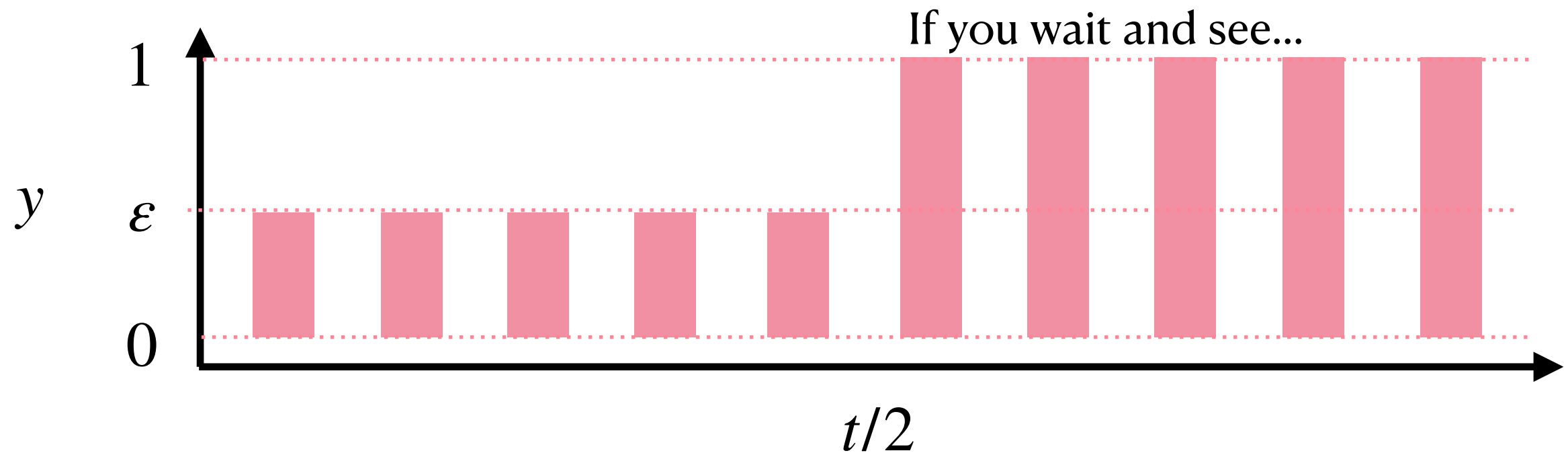
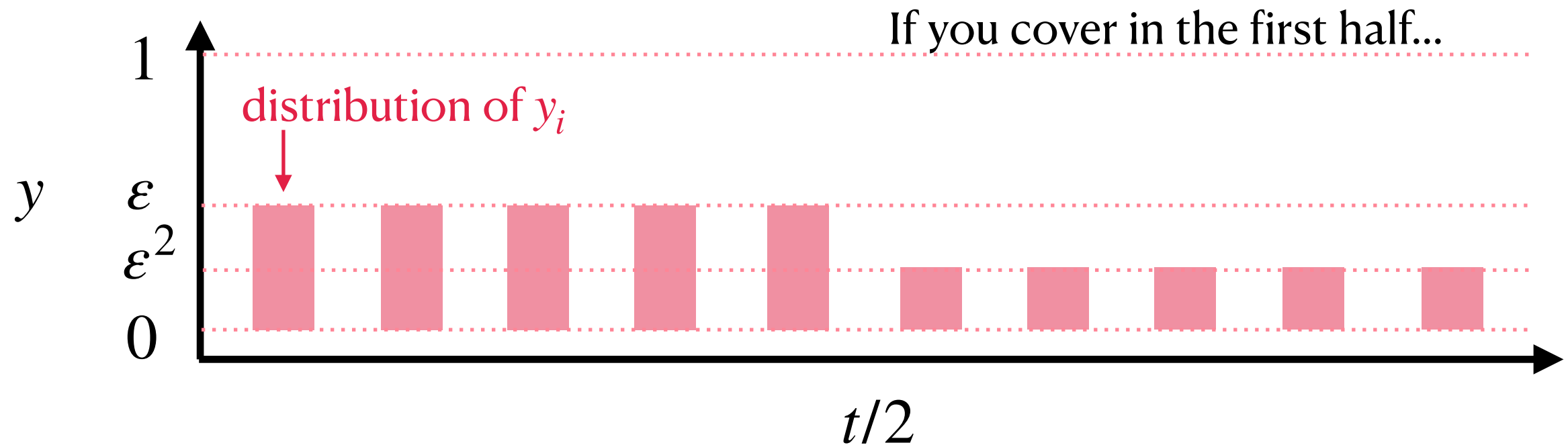
Only have $y_i \in [0,1]$ (all features x_i are the same)



Sneak peek: prediction set problem very different
from **regression/estimation** problem

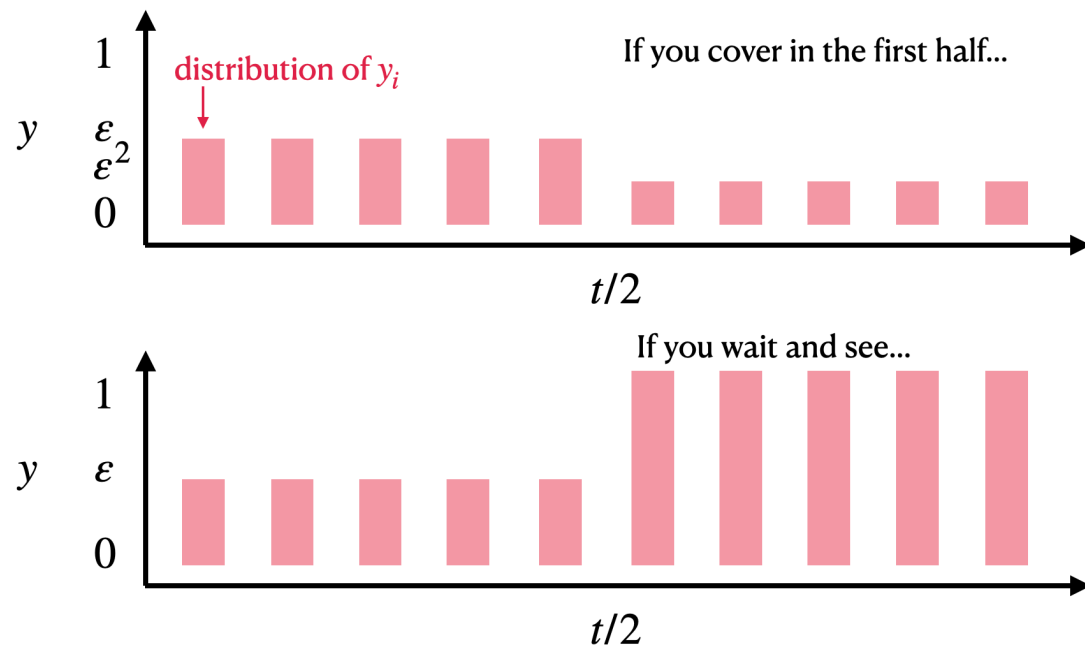
Hurdle

Goal: coverage $1 - \alpha = 0.5$



Have to lose in either **coverage** or **volume**

Are we toast?



Previous example:

For $1 - \alpha = 0.5$, can't simultaneously

- achieve non-trivial volume guarantee better than multiplicative $1/\epsilon$ -factor
- **and** capture $(1 - \alpha)$ -fraction of points

Relaxed objective:

Allow multiplicative approximation factors in

- **Volume:** compared to best interval that captures $(1 - \alpha)$ -fraction of points
- **Miscoverage:** number of points not covered, compared to target αT
 - Interesting for $\alpha < 1/2$

Updated Question: What are the Pareto-optimal **bicriteria approximations**?

Result: Arbitrary-order Sequences

$\text{Opt}_S(\alpha)$: volume of smallest interval in hindsight achieving coverage $1-\alpha$ on S

Informal Theorem [S. '25]

For a given scale lower bound $\varepsilon > 0$, multiplicative volume approximation $\mu > 3$, target miscoverage rate $\alpha \geq 0$, and time horizon T , we give a **deterministic algorithm** that on any sequence S of length T plays intervals of *maximum* volume

$$\leq \mu \max\{\text{Opt}_S(\alpha), \varepsilon\}, \text{ (efficiency)}$$

and makes number of mistakes bounded by

$$O\left(\frac{\log(1/\varepsilon)}{\log(\mu)}(\alpha T + 1)\right), \text{ (coverage)}$$

and this is near-optimal.

Tradeoff between coverage and efficiency, no **vanishing regret** possible!

Algorithm

Goal: optimize **volume** with respect to a **coverage** constraint

Intuition: Optimizing **coverage** with a **volume** constraint would be easy!

Convert **feasibility** $\longleftrightarrow$ **optimization**

Informal Algorithm:

Input: $\alpha < 1/2$, vol. approx. factor μ

$I_{\text{current}} \leftarrow [0,0]$

For day t :

- If I_{current} missed more than αT points seen so far:

- $I_t \leftarrow$ smallest interval that makes at most αT mistakes so far

- $I_{\text{current}} \leftarrow \mu I_t$

- Predict I_{current}

Optimize coverage with volume constraint (easy!)

Volume Approximation:

- Opt always feasible choice for I_t
- Never play intervals more than μ times bigger than Opt

Coverage Approximation:

- Only make αT mistakes every time we reset I_{current}
- Every time we reset I_{current} , it grows by factor $O(\mu)$

Result again: Arbitrary-order Sequences

Informal Theorem [S. '25]

We give a deterministic algorithm that on any sequence S of length T plays intervals of *maximum* volume

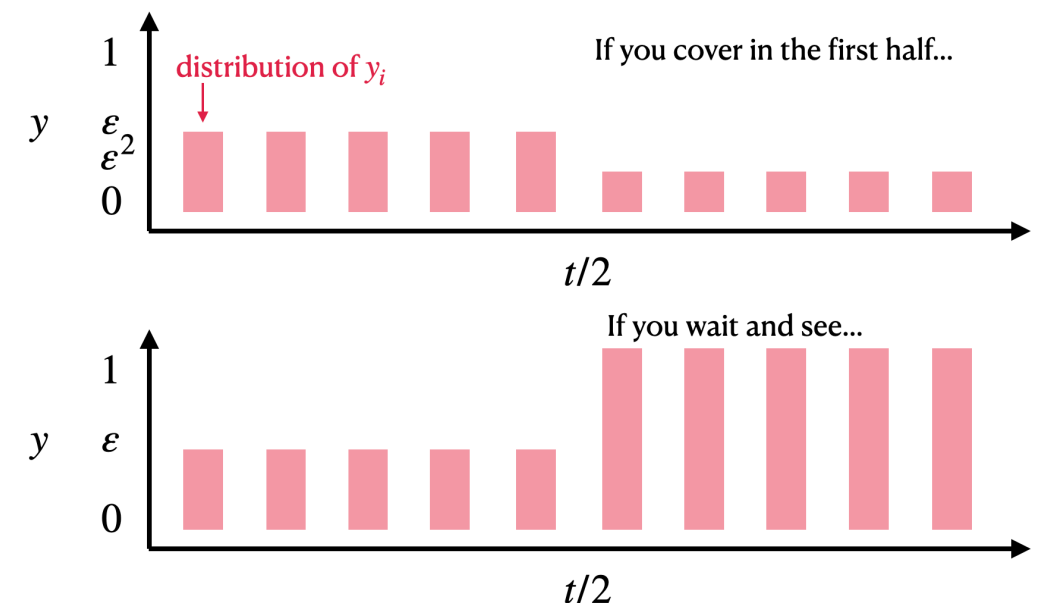
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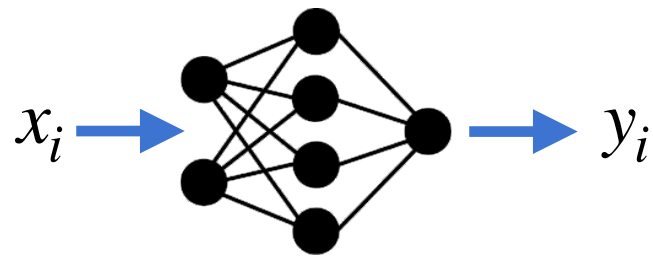
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Lower bound is a generalization of earlier example, with more stair steps



Implications for Conformal Prediction

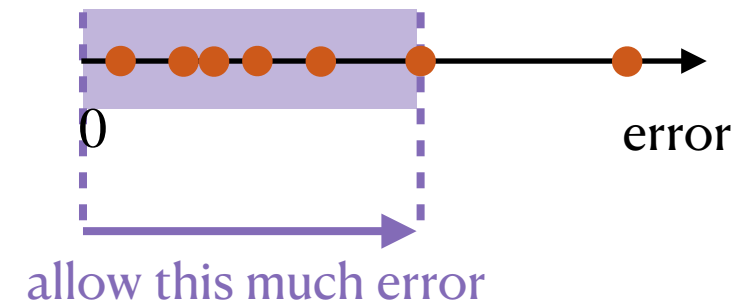
Standard Recipe for Conformal Prediction:



Train regression model

$$s(y, \hat{y}) = \|\hat{y} - y\|$$

Fix non-conformity score



Take $\approx 1 - \alpha$ quantile

Quantile regression achieves coverage approaching $1 - \alpha$ as $T \rightarrow \infty$

“vanishing regret”

Takeaway: this strategy achieves unboundedly bad volume approximations in the worst case!

Achieving small prediction set size = **efficient conformal prediction**
requires a different approach

Zoom Out: Results in this Work

Arbitrary order sequences:

- Must incur multiplicative-factor approximations in **volume** and **miscoverage**

Random order sequences:

- Can approach optimal **volume** and **miscoverage** simultaneously!
- Close to **standard conformal prediction** (vs. online conformal prediction)
- Almost the same algorithm! (Reset I_{current} more aggressively, for lower error rates)

Why settle for **almost**?

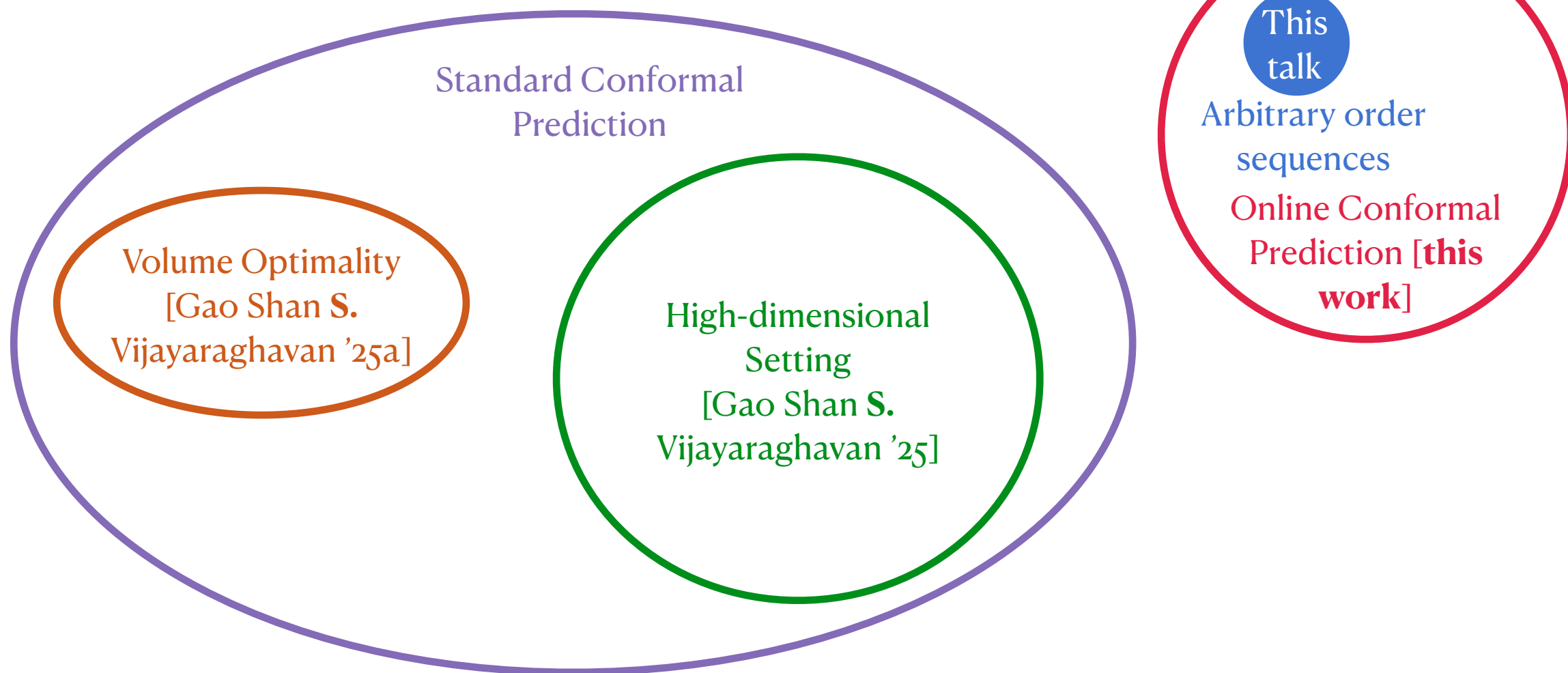
No best-of-both worlds:

- No single algorithm can be optimal for both arbitrary and random-order sequences
- Can design family of algorithms to achieve the optimal trade-off

Informs what kinds of guarantees we can hope for



(Zoom Out)²: Theory of Prediction Sets



Learning **prediction sets** is fundamentally different than other learning tasks like estimation, regression, and classification, and requires **new theory**

Open problems: almost everything! Come join us :)

Thanks!

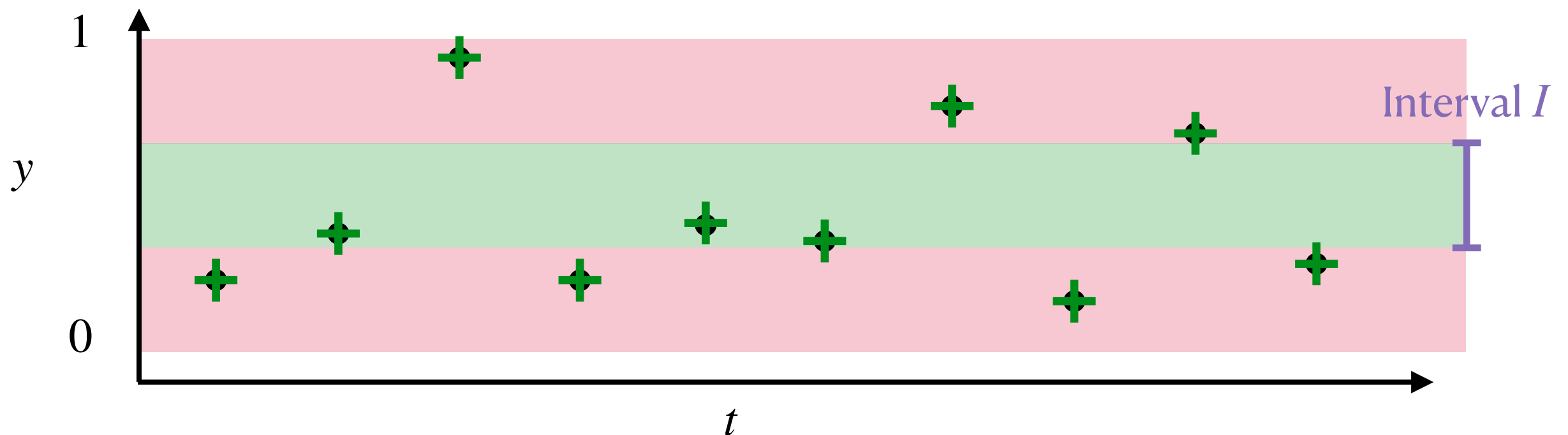
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Bonus Slides

Interpretation: Online Learning

Constrained Online Learning:

- Related to **binary classification** with hypothesis class of **intervals**
 - Think of every point y_i as labeled positive $\rightarrow (y_i, +)$
 - Learn best labeling of points as $+$ or $-$
- Minimize volume of intervals played, subject to classification error $\leq \alpha$



Takeaway: Unconstrained online learning admits **vanishing regret**,
but constrained version looks very different!

Algorithm: Volume

Goal: optimize **volume** with respect to a **coverage** constraint

Intuition: Optimizing **coverage** with a **volume** constraint would be easy!

Convert **feasibility** $\longleftrightarrow$ **optimization**

Informal Algorithm:

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Optimize coverage with volume constraint

Volume Approximation:

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Algorithm: Coverage

Coverage Approximation:

Informal Algorithm:

Input: $\alpha < 1/2$, vol. approx. factor μ

$I_{\text{current}} \leftarrow [0,0]$

For day t :

- If I_{current} missed more than αT points seen so far:
 - $I_t \leftarrow$ smallest interval that makes at most αT mistakes so far
 - $I_{\text{current}} \leftarrow \mu I_t$
- Predict I_{current}

- Each choice of I_{current} misses at most αT points
- Bound number of times we choose I_{current}
 - New I_t captures at least one point in the old I_t , and at least one point outside I_{current}
 - I_{current} grows by factor $\approx \mu/2$
- Bound number of iterations by
$$\frac{\log(1/\epsilon)}{\log(\mu)}$$

